Fiscal Year:	FY 2024	**Task Last Updated:**	FY 08/28/2025
PI Name:	Hindle, Allyson Ph.D.		
Project Title:	Can Humans Hibernate at Warm Temperatures?		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	TRISH--TRISH		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	89154-4004	**Congressional District:**	1
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2021 BRASH2101: Translational Research Institute for Space Health (TRISH) Biomedical Research Advances for Space Health
Start Date:	07/01/2022	**End Date:**	08/31/2024
No. of Post Docs:	2	**No. of PhD Degrees:**	
No. of PhD Candidates:	3	**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	1	**Monitoring Center:**	TRISH
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: Project extended to 08/31/2024 per TRISH (Ed., 5/16/24). NOTE: Period of performance changed per TRISH; original dates were 06/01/2022 - 5/31/2024.		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	van Breukelen, Frank Ph.D. (University of Nevada, Las Vegas)		
Grant/Contract No.:	NNX16AO69A-T0603		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	This project will determine the optimum body temperature for human hibernation, that maximizes energy savings while minimizing homeostatic disruption. Cold temperatures allow the greatest metabolic savings, but it is not known to what degree homeostatic control is maintained across temperatures. Natural hibernators such as bears retain warm body temperatures whereas ground squirrels allow body temperature to drop. Bears therefore sustain protein turnover during hibernation, and ground squirrels shut off homeostatic processes in the cold, thereby maintaining appropriate gene product representation when hibernating. Primitive tropical hibernators such as tenrecs maintain intermediate “warm” body temperatures, but appear to suffer the highest disruption of homeostatic processes of all species. One scenario to promote metabolic depression of spaceflight crews would be to provoke body temperature declines to ambient spacecraft temperatures; based on observations from natural hibernators, this is exactly the temperature at which we anticipate the maximum disruption of homeostasis would occur. This project will test the hypothesis that warm hibernation elicits the most homeostatic dysregulation in mammals, including humans, by tracking the control of gene expression (transcription) and protein synthesis (translation) in a ground squirrel model capable of hibernating to cold temperatures alongside human cells and tissues. This project will determine an optimum temperature for human “hibernation” in vitro that balances metabolic rate reduction with maintenance and recovery of homeostatic processes. As a result, we will produce a recommendation regarding the efficacy and safety of “hibernating” human crewmembers at ambient spacecraft temperatures. This project will use a parallel approach to investigate control of transcription and translation over a range of “hibernation” temperatures in a natural hibernator in vivo, alongside human cells and tissues in vitro. First, to determine whether homeostatic control is maintained in a natural model in cool, intermediate, and hot temperatures, we will hibernate golden mantled ground squirrels at 4, 12, 25, and 30°C to elicit different body temperatures. In addition to data on oxygen consumption (respirometry) and torpor patterns (body temperature telemetry) to determine efficacy of hibernation, we will measure differential gene expression (RNAseq from liver, heart, kidney, and brain) during torpor. Second, to identify the optimum temperature for human cells and organoids in vitro that best maintains homeostatic control, we will use experimental cooling to induce metabolic depression. We will quantify oxygen consumption reduction and mitochondrial function as indices of metabolic depression at 4, 12, 25, and 30°C and in parallel, evaluate transcription and translation rates by tracking incorporation of radioactive labels, as well as sequencing. If human cells cannot effectively maintain homeostasis at low temperatures, then either intervention is required, or high temperature hibernation is necessary. In our third aim, we will trial an intervention that promotes internal ribosome entry site (IRES)-mediated protein synthesis and track indicators of protein synthesis (polysomes). The Translational Research Institute for Space Health (TRISH) seeks innovation to reversibly manipulate metabolic and homeostatic processes on demand as a key spaceflight countermeasure, maximizing crew resilience and health, thus reducing mission risk. Temperature is a likely contributor to any strategy that will ultimately reduce metabolic processes. This project will examine the limits of temperature regulation to elicit metabolic depression in humans and natural hibernators and determine the tradeoffs between energy savings and regulation of homeostasis across varied temperatures, focusing on control of protein synthesis. This approach will take critical steps forward in developing temperature regulation (hypothermia), as well as stimulation of IRES-mediated protein synthesis as a countermeasure.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Research impacts are the development of primary cell types from the natural hibernator model, the golden mantled ground squirrel, along with associated protocols for proliferation and metabolic assays in these cells. These research tools will be available for future investigations by us and others. We have also obtained long-range genomic sequencing data from this species and built an assembly for the species, which will become publicly available once the genome annotation is complete. This project also contributes an important perspective to the field of hibernation biology, by evaluating hibernation across different temperatures. This is an atypical approach to study this physiological and biochemical phenomenon, which we expect will advance our understanding of metabolic tradeoffs associated with torpor.
Task Progress:	Innovation to reversibly manipulate metabolic and homeostatic processes on demand has been proposed as a key spaceflight countermeasure, maximizing crew resilience and health, thus reducing mission risk. Temperature is a likely contributor to any strategy that will ultimately reduce metabolic processes. This project examined the impact of temperature on elicited metabolic depression in humans and natural hibernators and investigated tradeoffs between energy savings and regulation of homeostasis across varied temperatures, focusing on control of protein synthesis. Aim 1 addressed metabolic savings, temperature dynamics and transcriptomics in a model hibernator. Our data in the golden mantled ground squirrel indicate that warm temperatures of hibernation result in the fewest transcriptomics adjustments, and are actively selected by hibernating animals. Torpor occurring at ambient temperatures of 30C result in more erratic/variable torpor dynamics, and body temperature does not significantly drop compared to active animals, however these animals still exhibit a significant (~55%) drop in metabolic rate. These data suggest that metabolic depression with minimal changes in body temperature is likely to be safest and will still result in energetic savings. Aim 2 addressed similar metrics but in human cells from 4 different tissues. All cells are able to tolerate temperatures as low as 4C, however cardiomyocytes become metabolically quiescent at 16C. Cells display significant reductions in basal rates of oxygen consumption even at 30C (the warmest temperature tested). In hepatocytes, maximal oxygen consumption and spare capacity increase at the lowest temperature tested (16C) suggesting that temperature may constrain metabolism, and reduced temperatures may impact metabolic savings in tissues in a distinct fashion. Our data also consistently shows that intermediate temperatures of hibernation evoke the most transcriptomic disruption, particularly in comparison to euthermic controls at 37C. We initially proposed an intervention of rapamycin treatment (Aim 3) to adjust transcriptomic profiles during rewarming, favoring IRES mediated protein production and improving human cell survival. As cell survival was not impacted by short-term cooling and rewarming in our hands, a more subtle metric is likely needed to evaluate this intervention. Because there are so few gene targets changing in expression levels between 37 and 30C treatments in human cells, we propose a more detailed in silico evaluation of the representation of genes with IRES secondary structure in the dataset. Natural hibernators such as bears retain warm body temperatures whereas ground squirrels allow body temperature to drop. Bears therefore sustain protein turnover during hibernation, and ground squirrels shut off homeostatic processes in the cold, thereby maintaining appropriate gene product representation when hibernating. Primitive tropical hibernators such as tenrecs maintain intermediate “warm” body temperatures, but appear to suffer the highest disruption of homeostatic processes of all species. One scenario to promote metabolic depression of spaceflight crews would be to provoke body temperature declines to ambient spacecraft

	temperatures–based on our observations, this is exactly the temperature at which we anticipate the maximum disruption of homeostasis would occur. Our data in both ground squirrels and humans suggest that the highest temperatures tested (30C) and sustainable with minimal transcriptomic disruption compared to baseline euthermic conditions and also result in appreciable energy savings.
Bibliography Type:	Description: (Last Updated: 08/28/2025)
Abstracts for Journals and Proceedings	Crippen T, Barth D, Carlson Z, van Breukelen F, Hindle AG. "Torpor in the golden-mantled ground squirrel: investigating transcriptomic dynamics across temperatures. " 2nd Annual American Physiology Summit, Long Beach, CA, April 2024. Abstracts. 2nd Annual American Physiology Summit, Long Beach, CA, April 2024. , Apr-2024
Abstracts for Journals and Proceedings	Silva Rubio C, Yankaway R, Smith GR, Hindle AG, van Breukelen F. "Hibernating golden-mantled ground squirrels (Spermophilus [Callospermophilus] lateralis) have high inter-individual variability in oxygen consumption rates with warmer ambient temperatures." 2nd Annual American Physiology Summit, Long Beach, CA, April 2024. Abstracts. 2nd Annual American Physiology Summit, Long Beach, CA, April 2024. , Apr-2024
Abstracts for Journals and Proceedings	Yankaway R, van Breukelen F, Hindle AG, Utz J. "Hibernating at high temperatures increases variability in arousal dynamics. " 2nd Annual American Physiology Summit, Long Beach, CA, April 2024. Abstracts. 2nd Annual American Physiology Summit, Long Beach, CA, April 2024. , Apr-2024
Awards	Hindle A. "UNLV Barrick Scholar, March 2024." Mar-2024